# CHAPTER 8

**8.1** A 4 mC charge has velocity $\mathbf{u} = 1.4\mathbf{a}_x - 3.2\mathbf{a}_y - \mathbf{a}_z$ m/s at point $P(2, 5, -3)$ in the presence of $\mathbf{E} = 2xyz\mathbf{a}_x + x^2z\mathbf{a}_y + x^2y\mathbf{a}_z$ V/m and $\mathbf{B} = y^2\mathbf{a}_x + z^2\mathbf{a}_y + x^2\mathbf{a}_z$ Wb/m$^2$. Find the force on the charge at $P$.

**8.2** An electron ($m = 9.11 \times 10^{-31}$ kg) moves in a circular orbit of radius $0.4 \times 10^{-10}$ m with an angular velocity of $2 \times 10^{16}$ rad/s. Find the centripetal force required to hold the electron.

**8.3** A 1 mC charge with velocity $10\mathbf{a}_x - 2\mathbf{a}_y + 6\mathbf{a}_z$ m/s enters a region where the magnetic flux density is $25\mathbf{a}_z$ Wb/m$^2$. (a) Calculate the force on the charge. (b) Determine the electric field intensity necessary to make the velocity of the charge constant.

**8.4** Assume an electric field intensity of 20 kV/m and a magnetic flux density of 5 $\mu$Wb/m$^2$ exist in a region. Find the ratio of the magnitudes of electric and magnetic forces on an electron that has attained a velocity of $0.5 \times 10^8$ m/s.

**8.5** A $-2$ mC charge starts at point $(0, 1, 2)$ with a velocity of $5\mathbf{a}_x$ m/s in a magnetic field $\mathbf{B} = 6\mathbf{a}_y$ Wb/m$^2$. Determine the position and velocity of the particle after 10 s, assuming that the mass of the charge is 1 gram. Describe the motion of the charge.

**8.6** By injecting an electron beam normally to the plane edge of a uniform field $B_o\mathbf{a}_z$, electrons can be dispersed according to their velocity as in Figure 8.34.

(a) Show that the electrons would be ejected out of the field in paths parallel to the input beam as shown.

(b) Derive an expression for the exit distance $d$ above the entry point.

**8.7** Two large conducting plates are 8 cm apart and have a potential difference 12 kV. A drop of oil with mass 0.4 g is suspended in space between the plates. Find the charge on the drop.

**8.8** A straight conductor 0.2 m long carries a current 4.5 A along $\mathbf{a}_x$. If the conductor lies in the magnetic field $\mathbf{B} = 2.5(\mathbf{a}_y + \mathbf{a}_z)$ mWb/m$^2$, calculate the force on the conductor.

**8.9** Determine $|\mathbf{B}|$ that will produce the same force on a charged particle moving at 140 m/s that an electric field of 12 kV/m produces.

**Prob. 8.1**

At P, x = 2, y = 5, z = -3

$$\boldsymbol{E} = 2(2)(5)(-3)\boldsymbol{a}_x + (2)^2(-3)\boldsymbol{a}_y + (2)^2(5)\boldsymbol{a}_z = -60\boldsymbol{a}_x - 12\boldsymbol{a}_y + 20\boldsymbol{a}_z$$

$$\boldsymbol{B} = (5)^2\boldsymbol{a}_x + (-3)^2\boldsymbol{a}_y + 2^2\boldsymbol{a}_z = 25\boldsymbol{a}_x + 9\boldsymbol{a}_y + 4\boldsymbol{a}_z$$

$$\boldsymbol{F} = Q(\boldsymbol{E} + \boldsymbol{u}\times\boldsymbol{B})$$

$$\boldsymbol{u}\times\boldsymbol{B} = \begin{vmatrix} 1.4 & -3.2 & -1 \\ 25 & 9 & 4 \end{vmatrix} = -3.8\boldsymbol{a}_x - 30.6\boldsymbol{a}_y + 92.6\boldsymbol{a}_z$$

$$\boldsymbol{E} + \boldsymbol{u}\times\boldsymbol{B} = (-60,-12,20) + (-3.8,-30.6,92.6) = (-63.8,-42.6,112.6)$$

$$\boldsymbol{F} = Q(\boldsymbol{E} + \boldsymbol{u}\times\boldsymbol{B}) = 4(\boldsymbol{E} + \boldsymbol{u}\times\boldsymbol{B}) \text{ mN}$$

$$= \underline{\underline{-255.2\boldsymbol{a}_x - 170.4\boldsymbol{a}_y + 450.4\boldsymbol{a}_z \text{ mN}}}$$

**Prob. 8.2**

$$F = m\omega^2 r = 9.11\times10^{-31}\times(2\times10^{16})^2(0.4\times10^{-10}) = \underline{\underline{14.576 \text{ nN}}}$$

**Prob. 8.3**

(a)

$$\boldsymbol{F} = Q(\boldsymbol{u}\times\boldsymbol{B}) = 10^{-3}\begin{vmatrix} 10 & -2 & 6 \\ 0 & 0 & 25 \end{vmatrix} = 10^{-3}(-50\boldsymbol{a}_x - 250\boldsymbol{a}_y)$$

$$= \underline{\underline{-0.05\boldsymbol{a}_x - 0.25\boldsymbol{a}_y \text{ N}}}$$

(b) Constant velocity implies that acceleration $\mathbf{a} = \mathbf{0}$.

$$\boldsymbol{F} = m\boldsymbol{a} = 0 = Q(\boldsymbol{E} + \boldsymbol{u}\times\boldsymbol{B})$$

$$\boldsymbol{E} = -\boldsymbol{u}\times\boldsymbol{B} = \underline{\underline{50\boldsymbol{a}_x + 250\boldsymbol{a}_y \text{ V/m}}}$$

**Prob. 8.4**

$$\mathbf{F}_e = q\mathbf{E}, \qquad \mathbf{F}_m = q\mathbf{u}\times\mathbf{B}$$

$$\frac{F_e}{F_m} = \frac{E}{uB} = \frac{20\times10^3}{0.5\times10^8\times5\times10^{-6}} = \underline{\underline{80}}$$

**Prob. 8.5**

$$m\mathbf{a} = Q\mathbf{u}\times\mathbf{B}$$

$$10^{-3}\mathbf{a} = -2\times10^{-3}\begin{vmatrix} u_x & u_y & u_z \\ 0 & 6 & 0 \end{vmatrix}$$

$$\frac{d}{dt}(u_x, u_y, u_z) = (12u_z, 0, -12u_x)$$

i.e. $\dfrac{du_x}{dt} = 12u_z$ (1

$$\frac{du_y}{dt} = 0 \rightarrow u_y = A_1 \quad (2$$

$$\frac{du_z}{dt} = -12u_x \quad (3$$

From (1) and (3),

$$\ddot{u}_x = 12\dot{u}_z = -144u_x$$

or

$$\ddot{u}_x + 144u_x = 0 \rightarrow u_x = c_1\cos 12t + c_2\sin 12t$$

From (1), $u_z = -c_1\sin 12t + c_2\cos 12t$

At t=0,

$u_x=5$, $u_y=0$, $u_z=0 \rightarrow A_1=0=c_2$, $c_1=5$

Hence,

$$\mathbf{u} = (5\cos 12t, 0, -5\sin 12t)$$

$$\mathbf{u}(t=10s) = (5\cos 120, 0, -5\sin 120) = \underline{\underline{4.071\mathbf{a}_x - 2.903\mathbf{a}_z}} \text{ m/s}$$

$$u_x = \frac{dx}{dt} = 5\cos 12t \rightarrow x = \tfrac{5}{12}\sin 12t + B_1$$

$$u_y = \frac{dy}{dt} = 0 \rightarrow y = B_2$$

$$u_z = \frac{dz}{dt} = -5\sin 12t \rightarrow z = \tfrac{5}{12}\cos 12t + B_3$$

At t=0, (x, y, z) = (0, 1, 2) $\rightarrow$ $B_1$=0, $B_2$=1, $B_3$=$\frac{19}{12}$

$$(x, y, z) = \left(\frac{5}{12}\sin 12t, 1, \frac{5}{12}\cos 12t + \frac{19}{12}\right) \qquad (4)$$

At t=10s,

$$(x, y, z) = \left(\frac{5}{12}\sin 120, 1, \frac{5}{12}\cos 120 + \frac{19}{12}\right) = \underline{(0.2419,\ 1,\ 1.923)}$$

By eliminating t from (4),

$x^2 + (z - 19/12)^2 = (5/12)^2$, $y = 1$ which is a circle in the y=1 plane with center at (0,1,19/12). The particle gyrates.

**Prob. 8.6**

(a) $m\boldsymbol{a} = -e(\boldsymbol{u} \times \boldsymbol{B})$

$$-\frac{m}{e}\frac{d}{dt}(u_x, u_y, u_z) = \begin{vmatrix} u_x & u_y & u_z \\ 0 & 0 & B_o \end{vmatrix} = u_y B_o \vec{a}_x - B_o u_x \vec{a}_y$$

$$\frac{du_z}{dt} = 0 \rightarrow u_z = c = 0$$

$$\frac{du_x}{dt} = -u_y \frac{B_o e}{m} = -u_y w, \text{ where w = } \frac{B_o e}{m}$$

$$\frac{du_y}{dt} = u_x w$$

Hence,

$$\ddot{u}_x = -w\dot{u}_y = -w^2 u_x$$

or $\ddot{u}_x + w^2 u_x = 0 \rightarrow u_x = A\cos wt + B\sin wt$

$$u_y = -\frac{\dot{u}_x}{w} = A\sin wt - B\cos wt$$

At t=0, $u_x = u_o$, $u_y = 0 \rightarrow A = u_o$, B=0

Hence,

$$u_x = u_o \cos wt = \frac{dx}{dt} \rightarrow x = \frac{u_o}{w}\sin wt + c_1$$

$$u_y = u_o \sin wt = \frac{dy}{dt} \rightarrow y = -\frac{u_o}{w}\cos wt + c_2$$

At t=0, x = 0 = y → $c_1=0$, $c_2=\frac{u_o}{w}$. Hence,

$$x=\frac{u_o}{w}\sin wt,\ y=\frac{u_o}{w}(1-\cos wt)$$

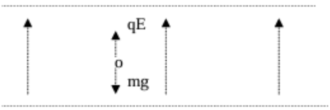
$$\frac{u^2_o}{w^2}(\cos^2 wt+\sin^2 wt)=\left(\frac{u_o}{w}\right)^2=x^2+(y-\frac{u_o}{w})^2$$

showing that the electron would move in a circle centered at $(0, \frac{u_o}{w})$. But since the field does not exist throughout the circular region, the electron passes through a semi-circle and leaves the field horizontally.

(b) d = twice the radius of the semi-circle

$$=\frac{2u_o}{w}=\frac{2u_o m}{B_o e}$$

**Prob. 8.7**




$$mg=qE \quad \longrightarrow \quad q=\frac{mg}{E}=\frac{0.4\times10^{-3}\times9.81}{1.5\times10^5}=\underline{\underline{26.67\ \text{nC}}}$$

**Prob. 8.8**

$$\boldsymbol{F}=\int I d\boldsymbol{l}\times\boldsymbol{B}=I\boldsymbol{L}\times\boldsymbol{B}$$

$$=4.5(0.2)\boldsymbol{a}_x\times(2.5)(\boldsymbol{a}_y+\boldsymbol{a}_z)10^{-3}=4.5(0.5)\begin{vmatrix}\boldsymbol{a}_x & \boldsymbol{a}_y & \boldsymbol{a}_z\\ 1 & 0 & 0\\ 0 & 1 & 1\end{vmatrix}\ \text{mN}$$

$$\boldsymbol{F}=\underline{2.25(-\boldsymbol{a}_y+\boldsymbol{a}_z)\ \text{mN}}$$

**Prob. 8.9**

$$qE=quB \quad \rightarrow \quad B=\frac{E}{u}=\frac{12\times10^3}{140}=\frac{1200}{14}=\underline{\underline{85.714\ \text{Wb/m}^2}}$$

**8.18** A 60-turn coil carries a current of 2 A and lies in the plane $x + 2y - 5z = 12$ such that the magnetic moment $\mathbf{m}$ of the coil is directed away from the origin. Calculate $\mathbf{m}$, assuming that the area of the coil is 8 cm$^2$.

**8.19** The earth has a magnetic moment of about $8 \times 10^{22}$ A · m$^2$ and its radius is 6370 km. Imagine that there is a loop around the equator and determine how much current in the loop would result in the same magnetic moment.

**8.20** A triangular loop is placed in the $x$-$z$ plane, as shown in Figure 8.39. Assume that a dc current $I = 2$ A flows in the loop and that $\mathbf{B} = 30\mathbf{a}_z$ m Wb/m exists in the region. Find the forces and torque on the loop.

**8.21** A loop with 50 turns and surface area of 12 cm$^2$ carries a current of 4 A. If the loop rotates in a uniform magnetic field of 100 mWb/m$^2$, find the torque exerted on the loop.

**8.22** High-current circuit breakers typically consist of coils that generate a magnetic field to blow out the arc formed when the contacts open. An arc 30 mm long carries a current of 520 A in a direction perpendicular to a magnetic flux density of 0.4 mWb/m$^2$. Determine the magnetic force on the arc.

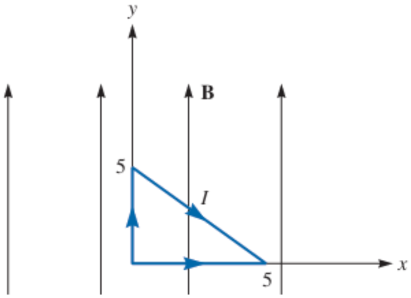



**FIGURE 8.39** For Problem 8.20.

## Prob. 8.18

$$f(x, y, z) = x + 2y - 5z - 12 = 0 \quad \longrightarrow \quad \nabla f = \boldsymbol{a}_x + 2\boldsymbol{a}_y - 5\boldsymbol{a}_z$$

$$\boldsymbol{a}_n = \frac{\nabla f}{|\nabla f|} = \frac{\boldsymbol{a}_x + 2\boldsymbol{a}_y - 5\boldsymbol{a}_z}{\sqrt{30}}$$

$$\boldsymbol{m} = NIS\boldsymbol{a}_n = 2\times 60\times 8\times 10^{-4}\,\frac{(\boldsymbol{a}_x + 2\boldsymbol{a}_y - 5\boldsymbol{a}_z)}{\sqrt{30}} = \underline{\underline{17.53\boldsymbol{a}_x + 35.05\boldsymbol{a}_y - 87.64\boldsymbol{a}_z \text{ mAm}}}$$

## Prob. 8.19

$$m = IS \quad \longrightarrow \quad I = \frac{m}{S} = \frac{m}{\pi r^2}$$

$$I = \frac{8\times 10^{22}}{\pi (6370\times 10^3)^2} = 6.275\times 10^8 = \underline{\underline{627.5 \text{ MA}}}$$

**Prob. 8.20**

Let $\boldsymbol{F} = \boldsymbol{F}_1 + \boldsymbol{F}_2 + \boldsymbol{F}_3$

$$\boldsymbol{F}_1 = \int I d\boldsymbol{l} \times \boldsymbol{B} = \int_5^0 2dx\boldsymbol{a}_x \times 30\boldsymbol{a}_z \ \text{mN}$$

$$= -60\boldsymbol{a}_y x \Big|_5^0 = 300\boldsymbol{a}_y \ \text{mN}$$

$$\boldsymbol{F}_2 = \int_0^5 2dy\boldsymbol{a}_y \times 30\boldsymbol{a}_z \ \text{mN}$$

$$= 60\boldsymbol{a}_x y \Big|_0^5 = 300\boldsymbol{a}_x \ \text{mN}$$

$$\boldsymbol{F}_3 = \int_0^5 2(dx\boldsymbol{a}_x + dz\boldsymbol{a}_z) \times 30\boldsymbol{a}_z \ \text{mN}$$

$$= 60(-\boldsymbol{a}_y)\, x \Big|_0^5 = -300\boldsymbol{a}_y \ \text{mN}$$

$$\boldsymbol{F} = \boldsymbol{F}_1 + \boldsymbol{F}_2 + \boldsymbol{F}_3 = 300\boldsymbol{a}_y + 300\boldsymbol{a}_x - 300\boldsymbol{a}_y \ \text{mN} = \underline{\underline{300\boldsymbol{a}_x \ \text{mN}}}$$

$$\boldsymbol{T} = \boldsymbol{m} \times \boldsymbol{B} = IS\boldsymbol{a}_n \times \boldsymbol{B} = 2(\frac{1}{2})(5)(5)\boldsymbol{a}_y \times 30\boldsymbol{a}_z 10^{-3} = \underline{\underline{0.75\boldsymbol{a}_x \ \text{N.m}}}$$

**Prob. 8.21**

For each turn, $\boldsymbol{T} = \boldsymbol{m} \times \boldsymbol{B}, \ \boldsymbol{m} = \text{IS}\boldsymbol{a}_n$

For N turns,

$$T = NISB = 50 \times 4 \times 12 \times 10^{-4} \times 100 \times 10^{-3} = \underline{\underline{24 \ \text{mNm}}}$$

**8.24** A block of iron ($\mu = 5000\mu_o$) is placed in a uniform magnetic field with 1.5 Wb/m$^2$. If iron consists of $8.5 \times 10^{28}$ atoms/m$^3$, calculate (a) the magnetization M, (b) the average magnetic moment.

**8.25** In a magnetic material, with $\chi_m = 6.5$, the magnetization is $\mathbf{M} = 24y^2\mathbf{a}_z$ A/m. Find $\mu_r$, $\mathbf{H}$, and $\mathbf{J}$ at $y = 2$ cm.

**8.26** In a ferromagnetic material ($\mu = 80\mu_o$), $\boldsymbol{B} = 20x\boldsymbol{a}_y$ mWb/m$^2$. Determine: (a) $\mu_r$, (b) $\chi_m$, (c) $\boldsymbol{H}$, (d) $\boldsymbol{M}$, (e) $\mathbf{J}_b$.

**Prob. 8.24**

(a) $\boldsymbol{M} = \chi_m \boldsymbol{H} = \chi_m \dfrac{\mathbf{B}}{\mu_o \mu}$

$$M = \frac{4999}{5000} \times \frac{1.5}{4\pi \times 10^{-7}} = \underline{\underline{1.193 \times 10^6 \text{ A/m}}}$$

(b) $\boldsymbol{M} = \dfrac{\sum_{k=1}^{N} \boldsymbol{m}_k}{\Delta v}$

If we assume that all $\boldsymbol{m}_k$ align with the applied **B** field,

$$M = \frac{Nm_k}{\Delta v} \quad \rightarrow \quad m_k = \frac{M}{N/\Delta v} = \frac{1.193 \times 10^6}{8.5 \times 10^{28}}$$

$$m_k = \underline{\underline{1.404 \times 10^{-23} \text{ A} \cdot \text{m}^2}}$$

**Prob. 8.25**

$$\mu_r = \chi_m + 1 = 6.5 + 1 = \underline{\underline{7.5}}$$

$$\boldsymbol{M} = \chi_m \boldsymbol{H} \quad \longrightarrow \quad \boldsymbol{H} = \frac{\boldsymbol{M}}{\chi_m} = \frac{24y^2}{6.5} \boldsymbol{a}_z$$

At y = 2cm,

$$\boldsymbol{H} = \frac{24 \times 4 \times 10^{-4}}{6.5} \boldsymbol{a}_z = \underline{\underline{1.477 \boldsymbol{a}_z \text{ mA/m}}}$$

$$\boldsymbol{J} = \nabla \times \boldsymbol{H} = \begin{vmatrix} \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ 0 & 0 & \dfrac{24y^2}{6.5} \end{vmatrix} = \frac{48y}{6.5} \mathbf{a}_x$$

At y=2cm,

$$\boldsymbol{J} = \frac{48 \times 2 \times 10^{-2}}{6.5} \mathbf{a}_x = \underline{\underline{0.1477 \boldsymbol{a}_x \text{ A/m}^2}}$$

**Prob. 8.26**

(a) $\mu = 80\mu_o \quad \rightarrow \quad \mu_r = 80$

(b) $\chi_m = \mu_r - 1 = 79$

(c) $\boldsymbol{H} = \dfrac{\boldsymbol{B}}{\mu} = \dfrac{20x\boldsymbol{a}_y 10^{-3}}{80(4\pi \times 10^{-7})} = 198.9x\boldsymbol{a}_y$ A/m

(d) $\boldsymbol{M} = \chi_m \boldsymbol{H} = 15.713x\boldsymbol{a}_y$ kA/m

(e) $\boldsymbol{J}_b = \nabla \times \boldsymbol{M} = \begin{vmatrix} \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ 0 & 15.713x & 0 \end{vmatrix} = 15.713\boldsymbol{a}_z$ kA/m

**8.30** Region 1, for which $\mu_1 = 2.5\mu_o$, is defined by $z < 0$, while region 2, for which $\mu_2 = 4\mu_o$, is defined by $z > 0$. If $\mathbf{B}_1 = 6\mathbf{a}_x - 4.2\mathbf{a}_y + 1.8\mathbf{a}_z$ mWb/m$^2$, find $\mathbf{H}_2$ and the angle $\mathbf{H}_2$ makes with the interface.

**8.31** In medium 1 ($z < 0$) $\mu_1 = 5\mu_o$, while in medium 2 ($z > 0$) $\mu_2 = 2\mu_o$. If $\boldsymbol{B}_1 = 4\boldsymbol{a}_x - 10\boldsymbol{a}_y + 12\boldsymbol{a}_z$ mWb/m$^2$, find $B_2$ and the energy density in medium 2.

**8.32** In region $x < 0$, $\mu = \mu_o$, a uniform magnetic field makes angle 42° with the normal to the interface. Calculate the angle the field makes with the normal in region $x > 0$, $\mu = 6.5\mu_o$.

**8.33** A current sheet with $\mathbf{K} = 12\mathbf{a}_y$ A/m is placed at $x = 0$, which separates region 1, $x < 0$, $\mu = 2\mu_o$ and region 2, $x > 0$, $\mu = 4\mu_o$. If $\mathbf{H}_1 = 10\mathbf{a}_x + 6\mathbf{a}_z$, A/m, find $\mathbf{H}_2$.

**8.34** Suppose space is divided into region 1 ($y < 0$, $\mu_1 = \mu_o\mu_{r1}$) and region 2 ($y < 0$, $\mu_2 = \mu_o\mu_{r2}$). If $\mathbf{H}_1 = \alpha\mathbf{a}_x + \beta\mathbf{a}_y + \delta\mathbf{a}_z$ A/m, find $\mathbf{H}_2$.

**Prob. 8.30**

$$\boldsymbol{B}_{2n} = \boldsymbol{B}_{1n} = 1.8\boldsymbol{a}_z$$

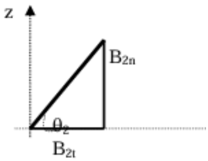
$$\boldsymbol{H}_{2t} = \boldsymbol{H}_{1t} \quad \longrightarrow \quad \frac{\boldsymbol{B}_{2t}}{\mu_2} = \frac{\boldsymbol{B}_{1t}}{\mu_1}$$

$$\boldsymbol{B}_{2t} = \frac{\mu_2}{\mu_1}\boldsymbol{B}_{1t} = \frac{4\mu_o}{2.5\mu_o}(6\boldsymbol{a}_x - 4.2\boldsymbol{a}_y) = 9.6\boldsymbol{a}_x - 6.72\boldsymbol{a}_y$$

$$\boldsymbol{B}_2 = \boldsymbol{B}_{2n} + \boldsymbol{B}_{2t} = 9.6\boldsymbol{a}_x - 6.72\boldsymbol{a}_y + 1.8\boldsymbol{a}_z \ \text{ mWb/m}^2$$

$$\boldsymbol{H}_2 = \frac{\boldsymbol{B}_2}{\mu_2} = \frac{10^{-3}(9.6, -6.72, 1.8)}{4\times 4\pi\times 10^{-7}}$$

$$= \underline{\underline{1,909.86\,\boldsymbol{a}_x - 1,336.9\boldsymbol{a}_y + 358.1\boldsymbol{a}_z \ \text{ A/m}}}$$




$$\tan\theta_2 = \frac{B_{2n}}{B_{2t}} = \frac{1.8}{\sqrt{9.6^2 + 6.72^2}} = 0.1536$$

$$\theta_2 = \underline{\underline{8.73^o}}$$

**Prob. 8.31**

$$\boldsymbol{B}_{2n} = \boldsymbol{B}_{1n} = 12\boldsymbol{a}_z$$

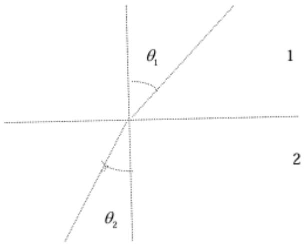
$$\boldsymbol{H}_{2t} = \boldsymbol{H}_{1t} \quad \longrightarrow \quad \frac{\boldsymbol{B}_{2t}}{\mu_2} = \frac{\boldsymbol{B}_{1t}}{\mu_1}$$

$$\boldsymbol{B}_{2t} = \frac{\mu_2}{\mu_1}\boldsymbol{B}_{1t} = \frac{2\mu_o}{5\mu_o}(4\boldsymbol{a}_x - 10\boldsymbol{a}_y) = 1.6\boldsymbol{a}_x - 4\boldsymbol{a}_y$$

$$\boldsymbol{B}_2 = \boldsymbol{B}_{2n} + \boldsymbol{B}_{2t} = \underline{\underline{1.6\boldsymbol{a}_x - 4\boldsymbol{a}_y + 12\boldsymbol{a}_z \text{ mWb/m}^2}}$$

$$w_2 = \frac{1}{2}\mu_2 H_2^3 = \frac{B_2^3}{2\mu_2} = \frac{(1.6^2 + 4^2 + 12^2)\times 10^{-6}}{2(2)(4\pi\times 10^{-7})} = \underline{\underline{32.34 \text{ J/m}^3}}$$

**Prob. 8.32**




$$\tan\theta_1 = \frac{H_{1t}}{H_{1n}}, \qquad \tan\theta_2 = \frac{H_{2t}}{H_{2n}}$$

But $\quad \mathrm{H}_{1t} = H_{2t}$

$$B_{1n} = B_{2t} \quad \rightarrow \quad \mu_1 H_{1n} = \mu_2 H_{2n} \quad \rightarrow \quad H_{2n} = \frac{\mu_1}{\mu_2}H_{1n}$$

$$\frac{H_{2t}}{H_{2n}} = \frac{H_{1t}}{\frac{1}{6.5}H_{1n}} = \frac{6.5 H_{1t}}{H_{1n}}$$

If $\quad \theta_1 = 42^o, \quad \tan 42^o = \dfrac{H_{1t}}{H_{1n}} \quad \rightarrow \quad \dfrac{H_{1t}}{H_{1n}} = 0.9004$

$$\tan\theta_2 = 6.5(0.9004) = 5.832 \quad \rightarrow \quad \underline{\underline{\theta_2 = 80.3^o}}$$

**Prob. 8.33**


$$(\boldsymbol{H}_1 - \boldsymbol{H}_2) \times \boldsymbol{a}_{n12} = \boldsymbol{K} \quad \to \quad \boldsymbol{H}_1 \times \mathbf{a}_x = \boldsymbol{K} + \boldsymbol{H}_2 \times \boldsymbol{a}_x$$

$$\boldsymbol{H}_1 \times \mathbf{a}_x = (10\boldsymbol{a}_x + 6\boldsymbol{a}_z) \times \boldsymbol{a}_x = 6\boldsymbol{a}_y$$

$$\boldsymbol{H}_2 \times \boldsymbol{a}_x = (H_{2x}, H_{2y}, H_{2z}) \times \boldsymbol{a}_x = H_{2z}\boldsymbol{a}_y - H_{2y}\boldsymbol{a}_z$$

$$6\boldsymbol{a}_y = 12\boldsymbol{a}_y + H_{2z}\boldsymbol{a}_y - H_{2y}\boldsymbol{a}_z$$

Equating components,

$$6 = 12 + H_{2z} \quad \to \quad H_{2z} = -6,$$

$$\mathrm{H}_{2y} = 0$$

Also, $\boldsymbol{B}_{1n} = \boldsymbol{B}_{2n} \quad \to \quad \mu_1 \boldsymbol{H}_{1n} = \mu_2 \boldsymbol{H}_{2n}$

$$\boldsymbol{H}_{2n} = \frac{\mu_1}{\mu_2}\boldsymbol{H}_{1n} = \frac{2\mu_o}{4\mu_o}(10\boldsymbol{a}_x) = 5\boldsymbol{a}_x$$

$$\underline{\underline{\boldsymbol{H}_2 = 5\boldsymbol{a}_x - 6\boldsymbol{a}_z \text{ A/m}}}$$

**Prob. 8.34**

$$\boldsymbol{H}_{2t} = \boldsymbol{H}_{1t} = \alpha\boldsymbol{a}_x + \delta\boldsymbol{a}_z$$

$$\boldsymbol{B}_{2n} = \boldsymbol{B}_{1n} \quad \longrightarrow \quad \mu_2 \boldsymbol{H}_{2n} = \mu_1 \boldsymbol{H}_{1n}$$

$$\boldsymbol{H}_{2n} = \frac{\mu_1}{\mu_2}\boldsymbol{H}_{1n} = \frac{\mu_{r1}}{\mu_{r2}}\beta\boldsymbol{a}_y$$

$$\boldsymbol{H} = \underline{\underline{\alpha\boldsymbol{a}_x + \frac{\mu_{r1}}{\mu_{r2}}\beta\boldsymbol{a}_y + \delta\boldsymbol{a}_z}}$$

---

**8.42** An air-filled toroid of square cross section has inner radius 3 cm, outer radius 5 cm, and height 2 cm. How many turns are required to produce an inductance of 45 $\mu$H?

**8.43** A wire of radius 2 mm is 40 m long. Calculate its inductance. Assume $\mu = \mu_o$.

**8.44** A coaxial cable has an internal inductance that is twice the external inductance. If the inner radius is 6.5 mm, calculate the outer radius.

**8.45** A hollow cylinder of radius $a = 2$ cm is 10 m long. Find the inductance of the cylinder. (See Table 8.3.)

**8.46** Show that the mutual inductance between the rectangular loop and the infinite line current of Figure 8.4 is

$$M_{12} = \frac{\mu b}{2\pi} \ln\left[\frac{a + \rho_o}{\rho_o}\right]$$

Calculate $M_{12}$ when $a = b = \rho_o = 1$ m.

**Prob. 8.42**

$$\rho_o = \frac{1}{2}(3+5) = 4\text{cm}$$

$a = 2$ cm

$$L = \frac{\mu_o N^2 a}{2\pi} \ln\left[\frac{2\rho_o + a}{2\rho_o - a}\right]$$

$$N^2 = \frac{2\pi L}{\mu_o a \ln\left[\frac{2\rho_o + a}{2\rho_o - a}\right]} = \frac{2\pi(45\times10^{-6})}{4\pi\times10^{-7}(2\times10^{-2})\ln\left(\frac{8+2}{8-2}\right)} = 22,023.17$$

$\underline{\underline{N = 148.4 \text{ or } 148}}$

**Prob. 8.43**

$$L = \frac{\mu_o \ell}{8\pi} = \frac{4\pi\times10^{-7}(40)}{8\pi} = 20\times10^{-7} = \underline{\underline{2\ \mu\text{H}}}$$

**Prob. 8.44**

$$L_{in} = \frac{\mu_o \ell}{8\pi}, \qquad L_{ext} = \frac{\mu_o \ell}{2\pi}\ln(b/a)$$

If $L_{in} = 2L_{ext} \longrightarrow \frac{\mu_o \ell}{8\pi} = \frac{\mu_o \ell}{\pi}\ln(b/a)$

$$\ln(b/a) = \frac{1}{8} \qquad \frac{b}{a} = e^{1/8} = 1.1331$$

$b = 1.1331a = \underline{\underline{7.365 \text{ mm}}}$

**Prob. 8.45**

$$L = \frac{\mu_o \ell}{2\pi}\left[\ln\frac{2\ell}{a} - 1\right] = \frac{4\pi\times10^{-7}(10)}{2\pi}\left[\ln\frac{2\times10}{2\times10^{-2}} - 1\right] = 2\times10^{-6}(\ln 1000 - 1)$$

$$= 2(5.908)\ \mu H = \underline{\underline{11.82\ \mu\text{H}}}$$

**Prob. 8.46**

$$\psi_{12} = \int \mathbf{B}_1 \bullet d\mathbf{S} = \int_{\rho=\rho_o}^{\rho_o + a} \int_{z=0}^{b} \frac{\mu_o I}{2\pi\rho} dz d\rho = \frac{\mu_o I b}{2\pi} \ln\frac{a + \rho_o}{\rho_o}$$

$$M_{12} = \frac{N\psi_{12}}{I} = \frac{N\mu_o b}{2\pi}\ln\frac{a+\rho_o}{\rho_o}$$

For N = 1,

$$M_{12} = \frac{\psi_{12}}{I_1} = \frac{\mu_o b}{2\pi}\ln\frac{a+\rho_o}{\rho_o}$$

$$= \frac{4\pi\times10^{-7}}{2\pi}(1)\ln 2 = \underline{\underline{0.1386\mu\,\text{H}}}$$

---

**8.50** In a certain region for which $\chi_m = 19$,

$$\mathbf{H} = 5x^2yz\mathbf{a}_x + 10xy^2z\mathbf{a}_y - 15xyz^2\mathbf{a}_z \text{ A/m}$$

How much energy is stored in $0 < x < 1, 0 < y < 2, -1 < z < 2$?

**8.51** The magnetic field in a material space ($\mu = 15\mu_o$) is given by

$$\mathbf{B} = 4\mathbf{a}_x + 12\mathbf{a}_y \text{ mWb/m}^2$$

Calculate the energy stored in region $0< x <2, 0 < y < 3, 0 < z < 4$.

**Prob. 8.50**

$$\mu_r = \chi_m + 1 = 20$$

$$w_m = \frac{1}{2} B_1 \cdot H_1 = \frac{1}{2}\mu \mathbf{H} \cdot \mathbf{H}$$

$$= \frac{1}{2}\mu\left(25x^4y^2z^2 + 100x^2y^4z^2 + 225x^2y^2z^4\right)$$

$$W_m = \int w_m dv$$

$$= \frac{1}{2}\mu\left[25\int_0^1 x^4dx\int_0^2 y^2dy\int_{-1}^2 z^2dz + 100\int_0^1 x^2dx\int_0^2 y^4dy\int_{-1}^2 z^2dz\right.$$

$$\left. + 225\int_0^1 x^2dx\int_0^2 y^2dy\int_{-1}^2 zdz\right]$$

$$= \frac{25\mu}{2}\left[\left.\frac{x^5}{5}\right|_0^1 \left.\frac{y^3}{3}\right|_0^2 \left.\frac{z^3}{3}\right|_{-1}^2 + 4\left.\frac{x^3}{3}\right|_0^1 \left.\frac{y^5}{5}\right|_0^2 \left.\frac{z^3}{3}\right|_{-1}^2\right.$$

$$\left. + 9\left.\frac{x^3}{3}\right|_0^1 \left.\frac{y^3}{3}\right|_0^2 \left.\frac{z^5}{5}\right|_{-1}^2\right]$$

$$= \frac{25\mu}{2}\left(\frac{1}{5}\cdot\frac{8}{3}\cdot\frac{9}{3} + \frac{4}{3}\cdot\frac{32}{3}\cdot\frac{9}{3} + \frac{9}{3}\cdot\frac{8}{3}\cdot\frac{33}{5}\right)$$

$$= \frac{25}{2}\times 4\pi\times 10^{-7}\times 20\times\frac{3600}{45}$$

$$W_m = \underline{\underline{25.13\,\text{mJ}}}$$

**Prob. 8.51**

$$W = \frac{1}{2}\int_v \mathbf{B} \bullet \mathbf{H} dv = \int_v \frac{|\mathbf{B}|^2}{2\mu} dv = \frac{1}{2(15)4\pi\times 10^{-7}} \int_{z=0}^{4}\int_{y=0}^{3}\int_{x=0}^{2} (4^2 + 12^2)10^{-6}\,dxdydz$$

$$= \frac{10^{-6}}{(30)4\pi\times 10^{-7}}(16+144)(2)(3)(4) = \frac{320}{\pi} = \underline{\underline{101.86\text{ J}}}$$

# CHAPTER 7

**7.1** (a) State Biot–Savart's law.

(b) The $y$- and $z$-axes, respectively, carry filamentary currents 10 A along $\mathbf{a}_y$ and 20 A along $-\mathbf{a}_z$. Find **H** at $(-3, 4, 5)$.

**7.2** A long, straight wire carries current 2A. Calculate the distance from the wire when the magnetic field strength is 10 mA/m.

**7.3** Two infinitely long wires, placed parallel to the z-axis, carry currents 10 A in opposite directions as shown in Figure 7.28. Find **H** at point *P*.

**7.4** Two current elements $I_1dl_1 = 4 \times 10^{-5}\,\mathbf{a}_x$ A.m at (0, 0, 0) and $I_2dl_2 = 6 \times 10^{-5}\,\mathbf{a}_y$ A.m at (0, 0, 1) are in free space. Find **H** at (3, 1, –2).

**7.6** Consider $AB$ in Figure 7.29 as part of an electric circuit. Find **H** at the origin due to $AB$.

**7.7** Line $x = 0, y = 0, 0 \le z \le 10$ m carries current 2 A along $\mathbf{a}_z$. Calculate **H** at points

(a) $(5, 0, 0)$
(b) $(5, 5, 0)$
(c) $(5, 15, 0)$
(d) $(5, -15, 0)$

**Prob. 7.1**

(a) See text

(b) Let $\mathbf{H} = \mathbf{H}_y + \mathbf{H}_z$

For $\boldsymbol{H}_z = \dfrac{I}{2\pi\rho}\boldsymbol{a}_\phi \qquad \rho = \sqrt{(-3)^2 + 4^2} = 5$

$$\mathbf{a}_\phi = -\mathbf{a}_z \times \frac{(-3\mathbf{a}_x + 4\mathbf{a}_y)}{5} = \frac{(3\mathbf{a}_y - 4\mathbf{a}_x)}{5}$$

$$\mathbf{H}_z = \frac{20}{2\pi(25)}(4\mathbf{a}_x + 3\mathbf{a}_y) = 0.5093\mathbf{a}_x + 0.382\mathbf{a}_y$$

For $\mathbf{H}_y = \dfrac{\mathbf{I}}{2\pi\rho}\mathbf{a}_\phi, \; \rho = \sqrt{(-3)^2 + 5^2} = \sqrt{34}$

$$\boldsymbol{a}_\phi = \boldsymbol{a}_y \times \frac{(-3\boldsymbol{a}_x + 5\boldsymbol{a}_z)}{\sqrt{34}} = \frac{3\boldsymbol{a}_z + 5\boldsymbol{a}_x}{\sqrt{34}}$$

$$H_y = \frac{10}{2\pi(34)}(5\mathbf{a}_x + 3\mathbf{a}_z) = 0.234\mathbf{a}_x + 0.1404\mathbf{a}_z$$

$$\mathbf{H} = \mathbf{H}_y + \mathbf{H}_z$$

$$= \underline{\underline{0.7433\mathbf{a}_x + 0.382\mathbf{a}_y + 0.1404\mathbf{a}_z \text{ A/m}}}$$

**Prob. 7.2**

$$H = \frac{I}{2\pi\rho} \quad \rightarrow \quad \rho = \frac{I}{2\pi H} = \frac{2}{2\pi(10\times10^{-3})} = \frac{100}{\pi} = \underline{\underline{31.83 \text{ m}}}$$

**Prob. 7.3**

Let $\quad \boldsymbol{H} = \boldsymbol{H}_1 + \boldsymbol{H}_2$

where $\boldsymbol{H}_1$ and $\boldsymbol{H}_2$ are respectively due to the lines located at (0,0) and (0,5).

$$\boldsymbol{H}_1 = \frac{I}{2\pi\rho}\boldsymbol{a}_\phi, \quad \rho = 5, \quad \boldsymbol{a}_\phi = \boldsymbol{a}_\ell \times \boldsymbol{a}_\rho = \boldsymbol{a}_z \times \boldsymbol{a}_x = \boldsymbol{a}_y$$

$$\boldsymbol{H}_1 = \frac{10}{2\pi(5)}\boldsymbol{a}_y = \frac{\boldsymbol{a}_y}{\pi}$$

$$\boldsymbol{H}_2 = \frac{I}{2\pi\rho}\boldsymbol{a}_\phi, \quad \rho = 5\sqrt{2}, \quad \boldsymbol{a}_\phi = \boldsymbol{a}_\ell \times \boldsymbol{a}_\rho, \boldsymbol{a}_\ell = -\boldsymbol{a}_z$$

$$\boldsymbol{a}_\rho = \frac{5\boldsymbol{a}_x - 5\boldsymbol{a}_y}{5\sqrt{2}} = \frac{\boldsymbol{a}_x - \boldsymbol{a}_y}{\sqrt{2}}$$

$$\boldsymbol{a}_\phi = -\boldsymbol{a}_z \times \left(\frac{\boldsymbol{a}_x - \boldsymbol{a}_y}{\sqrt{2}}\right) = \frac{-\boldsymbol{a}_x - \boldsymbol{a}_y}{\sqrt{2}}$$

$$\boldsymbol{H}_2 = \frac{10}{2\pi 5\sqrt{2}}\left(\frac{-\boldsymbol{a}_x - \boldsymbol{a}_y}{\sqrt{2}}\right) = \frac{1}{2\pi}(-\boldsymbol{a}_x - \boldsymbol{a}_y)$$

$$\boldsymbol{H} = \boldsymbol{H}_1 + \boldsymbol{H}_2 = \frac{\boldsymbol{a}_y}{\pi} + \frac{1}{2\pi}(-\boldsymbol{a}_x - \boldsymbol{a}_y) = \underline{\underline{-0.1592\boldsymbol{a}_x + 0.1592\boldsymbol{a}_y}}$$

**7.10** Find **H** at the center *C* of an equilateral triangular loop of side 4 m carrying 5 A of current as in Figure 7.31.

**Prob. 7.10**

For the side of the loop along y-axis,

$$\boldsymbol{H}_1 = \frac{\mathrm{I}}{4\pi\rho}(\cos\alpha_2 - \cos\alpha_1)\ \boldsymbol{a}_\phi$$

where $\boldsymbol{a}_\phi = -\boldsymbol{a}_x, \ \rho = 2\tan 30^\circ = \frac{2}{\sqrt{3}}, \ \alpha_2 = 30^\circ, \ \alpha_1 = 150^\circ$

$$\boldsymbol{H}_1 = \frac{5}{4\pi}\frac{\sqrt{3}}{2}\left(\cos 30^\circ - \cos 150^\circ\right)(-\boldsymbol{a}_x) = -\frac{15}{8\pi}\boldsymbol{a}_x$$

$$\boldsymbol{H} = 3\boldsymbol{H}_1 = -1.79\boldsymbol{a}_x \text{ A/m}$$

**7.16** Two identical loops are parallel and separated by distance $d$ as shown in Figure 7.35. Calculate **H** at $(0, 0, d)$ assuming that $a = 3$ cm, $d = 4$ cm, and $I = 10$ A.

**7.17** A solenoid of radius 4 mm and length 2 cm has 150 turns/m and carries a current of 500 mA. Find (a) $|\mathbf{H}|$ at the center, (b) $|\mathbf{H}|$ at the ends of the solenoid.

**Prob. 7.16**

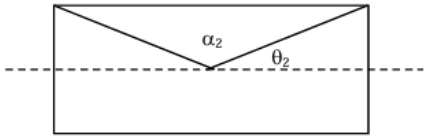
From Example 7.3,

$$\boldsymbol{H}_1 = \frac{I}{2a}\boldsymbol{a}_z, \qquad \boldsymbol{H}_2 = \frac{Ia^2}{2[a^2+d^2]^{3/2}}\boldsymbol{a}_z$$

$$\boldsymbol{H} = \boldsymbol{H}_1 + \boldsymbol{H}_2 = \frac{I}{2}\left[\frac{1}{a} + \frac{a^2}{[a^2+d^2]^{3/2}}\right]\boldsymbol{a}_z = \frac{10}{2}\left[\frac{1}{3\times10^{-2}} + \frac{3^2}{[3^2+4^2]^{3/2}(10^{-2})}\right]\boldsymbol{a}_z$$

$$= 500\left[\frac{1}{3} + \frac{9}{125}\right]\boldsymbol{a}_z = \underline{\underline{202.67\boldsymbol{a}_z \text{ A/m}}}$$

**Prob. 7.17**

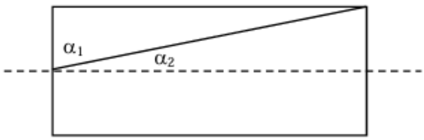



$$|\boldsymbol{H}| = \frac{nI}{2}(\cos\theta_2 - \cos\theta_1)$$

$$\cos\theta_2 = -\cos\theta_1 = \frac{\ell/2}{\left(a^2 + \ell^2/4\right)^{1/2}}$$

$$|\boldsymbol{H}| = \frac{nI\ell}{2\left(a^2 + \ell^2/4\right)^{1/2}} = \frac{0.5\times150\times2\times10^{-2}}{2\times10^{-3}\times\sqrt{4^2+10^2}} = \underline{\underline{69.63 \text{ A/m}}}$$

(b)




$$\alpha_1 = 90^\circ, \quad \tan\theta_2 = \frac{a}{b} = \frac{4}{20} = 0.2 \rightarrow \theta_2 = 11.31^\circ$$

$$|\boldsymbol{H}| = \frac{nI}{2}\cos\theta_2 = \frac{150\times0.5}{2}\cos 11.31^\circ = \underline{\underline{36.77 \text{ A/m}}}$$

**7.34** In free space, the magnetic flux density is

$$\mathbf{B} = y^2\mathbf{a}_x + z^2\mathbf{a}_y + x^2\mathbf{a}_z \text{ Wb/m}^2$$

(a) Show that **B** is a magnetic field
(b) Find the magnetic flux through $x = 1, 0 < y < 1, 1 < z < 4$.
(c) Calculate **J**.
(d) Determine the total magnetic flux through the surface of a cube defined by $0 < x < 2, 0 < y < 2, 0 < z < 2$.

**Prob. 7.34**

(a) $\nabla \bullet \boldsymbol{B} = \dfrac{\partial B_x}{\partial x} + \dfrac{\partial B_y}{\partial y} + \dfrac{\partial B_z}{\partial z} = 0$

showing that **B** satisfies Maxwell's equation.

(b) $d\boldsymbol{S} = dydz\boldsymbol{a}_x$

$$\Psi = \int \boldsymbol{B} \bullet d\boldsymbol{S} = \int_{z=1}^{4} \int_{y=0}^{1} y^2 dydz = \frac{y^3}{3}\Big|_0^1 (z)\Big|_1^4 = \underline{\underline{1 \text{ Wb}}}$$

(c) $\nabla \times \mathbf{H} = \mathbf{J} \quad \longrightarrow \quad \boldsymbol{J} = \nabla \times \dfrac{\boldsymbol{B}}{\mu_o}$

$$\nabla \times \boldsymbol{B} = \begin{vmatrix} \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ y^2 & z^2 & x^2 \end{vmatrix} = -2z\boldsymbol{a}_x - 2x\boldsymbol{a}_y - 2y\boldsymbol{a}_z$$

$$\underline{\underline{\boldsymbol{J} = -\frac{2}{\mu_o}(z\boldsymbol{a}_x + x\boldsymbol{a}_y + y\boldsymbol{a}_z) \text{ A/m}^2}}$$

(d)
Since $\nabla \bullet \boldsymbol{B} = 0$,

$$\Psi = \oint_S \boldsymbol{B} \bullet d\boldsymbol{S} = \int_v \nabla \bullet \boldsymbol{B} dv = \underline{\underline{0}}$$

**7.37** In free space, $\mathbf{B} = \dfrac{20}{\rho} \sin^2 \phi \mathbf{a}_z$ Wb/m². Determine the magnetic flux crossing the strip $z = 0, 1 < \rho < 2$ m, $0 < \phi < \pi/4$.

**7.40** Consider the following arbitrary fields. Find out which of them can possibly represent an electrostatic or magnetostatic field in free space.

(a) $\mathbf{A} = y \cos ax\mathbf{a}_x + (y + e^{-x})\mathbf{a}_z$

(b) $\mathbf{B} = \dfrac{20}{\rho}\mathbf{a}_\rho$

(c) $\mathbf{C} = r^2 \sin\theta\, \mathbf{a}_\phi$

**7.41** Reconsider Problem 7.40 for the following fields.

(a) $\mathbf{D} = y^2 z\mathbf{a}_x + 2(x+1)yz\mathbf{a}_y - (x+1)z^2\mathbf{a}_z$

(b) $\mathbf{E} = \dfrac{(z+1)}{\rho}\cos\phi\,\mathbf{a}_\rho + \dfrac{\sin\phi}{\rho}\mathbf{a}_z$

(c) $\mathbf{F} = \dfrac{1}{r^2}(2\cos\theta\,\mathbf{a}_r + \sin\theta\,\mathbf{a}_\theta)$

**Prob. 7.40**

(*a*) $\nabla \square \boldsymbol{A} = -ya\sin ax \neq 0$

$$\nabla \times \boldsymbol{A} = \begin{vmatrix} \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ y\cos ax & 0 & y+e^{x} \end{vmatrix}$$

$$= \boldsymbol{a}_x + e^{-x}\boldsymbol{a}_y - \cos ax\boldsymbol{a}_z \neq \boldsymbol{0}$$

***A*** is neither electrostatic nor magnetostatic field

(*b*) $\nabla\cdot\boldsymbol{B} = \dfrac{1}{\rho}\dfrac{\partial}{\partial\rho}(\rho B_\rho) = \dfrac{1}{\rho}\dfrac{\partial}{\partial\rho}(20) = 0$

$\nabla\times\boldsymbol{B} = \boldsymbol{0}$

**B** can be **E**-field in a charge-free region.

(*c*) $\nabla\cdot\mathbf{C} = \dfrac{1}{r\sin\theta}\dfrac{\partial}{\partial\phi}(r^2\sin\theta) = 0$

$\nabla\times\boldsymbol{C} = \dfrac{1}{r\sin\theta}\dfrac{\partial}{\partial\theta}(r^2\sin^2\theta)\boldsymbol{a}_r - \dfrac{1}{r}\dfrac{\partial}{\partial r}(r^3\sin\theta)\boldsymbol{a}_\theta \neq \boldsymbol{0}$

***C*** is possibly ***H*** field.

**Prob. 7.41**

(*a*) $\nabla\cdot\boldsymbol{D} = 0$

$$\nabla\times\boldsymbol{D} = \begin{vmatrix} \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ y^2 z & 2(x+1)yz & -(x+1)z^2 \end{vmatrix}$$

$$= 2(x+1)y\boldsymbol{a}_x + \ldots \neq \boldsymbol{0}$$

***D*** is possibly a magnetostatic field.

(*b*) $\nabla\cdot\mathbf{E} = \dfrac{1}{\rho}\dfrac{\partial}{\partial\rho}((z+1)\cos\phi) + \dfrac{\partial}{\partial z}\left(\dfrac{\sin\phi}{\rho}\right) = 0$

$\nabla\times\boldsymbol{E} = \dfrac{1}{\rho^2}\cos\theta\,\boldsymbol{a}_\rho + \ldots \neq \boldsymbol{0}$

***E*** could be a magnetostatic field.

(*c*) $\nabla\cdot\boldsymbol{F} = \dfrac{1}{r^2}\dfrac{\partial}{\partial r}(2\cos\theta) + \dfrac{1}{r\sin\theta}\dfrac{\partial}{\partial\theta}\left(\dfrac{\sin\theta}{r^2}\right) \neq 0$

$\nabla\times\boldsymbol{F} = \dfrac{1}{r}\left[\dfrac{\partial}{\partial r}(r^{-1}\sin\theta) + \dfrac{2\sin\theta}{r^2}\right]\boldsymbol{a}_\theta \neq \boldsymbol{0}$

***F*** can be neither electrostatic nor magnetostatic field.

# CHAPTER 6

**6.13** Two conducting coaxial cylinders are located at $\rho = 1$ cm and $\rho = 1.5$ cm. The inner conductor is maintained at 50 V while the outer one is grounded. If the cylinders are separated by a dielectric material with $\varepsilon = 4\varepsilon_o$, find the surface charge density on the inner conductor.

**6.14** Consider the conducting plates shown in Figure 6.30. If $V(z = 0) = 0$ and $V(z = 2\text{ mm}) = 50$ V, determine $V$, **E**, and **D** in the dielectric region ($\varepsilon_r = 1.5$) between the plates and $\rho_S$ on the plates.

**6.15** The cylindrical structure whose cross section is in Figure 6.31 has inner and outer radii of 5 mm and 15 mm, respectively. If $V(\rho = 5\text{ mm}) = 100$ V and $V(\rho = 15\text{ mm}) = 0$ V, calculate $V$, **E**, and **D** at $\rho = 10$ mm and $\rho_S$ on each plate. Take $\varepsilon_r = 2.0$.

**6.16** In cylindrical coordinates, $V = 50$ V on plane $\phi = \pi/2$ and $V = 0$ on plane $\phi = 0$. Assuming that the planes are insulated along the $z$-axis, determine **E** between the planes.

**Prob. 6.13**

$$\nabla^2 V = \frac{1}{\rho}\frac{d}{d\rho}\left(\rho\frac{dV}{d\rho}\right) = 0 \quad \rightarrow \quad V = A\ln\rho + B$$

Let a = 1 cm, b = 1.5 cm, $V_o = 50$V

$$V(\rho = b) = 0 \quad \rightarrow \quad 0 = A\ln b + B \text{ or } B = -A\ln b$$

$$V(\rho = a) = V_o \quad \rightarrow \quad V_o = A\ln a - A\ln b = A\ln\frac{a}{b} \text{ or } A = \frac{V_o}{\ln\frac{a}{b}}$$

$$V = A\ln\rho - A\ln b = A\ln\frac{\rho}{b}$$

$$\boldsymbol{E} = -\nabla V = -\frac{dV}{d\rho}\boldsymbol{a}_\rho = -\frac{A}{\rho}\boldsymbol{a}_\rho = -\frac{V_o}{\rho\ln\frac{a}{b}}\boldsymbol{a}_\rho$$

$$\rho_s = D_n = \varepsilon E_n = \frac{\varepsilon_o\varepsilon_r V_o}{a\ln\frac{b}{a}} = \frac{50(4)\frac{10^{-9}}{36\pi}}{10^{-2}\ln 1.5} = \frac{400(50)}{36\pi\ln 1.5}\text{ nC/m}^2 = \underline{\underline{436.14\text{ nC/m}^2}}$$

**Prob. 6.14**

$$\nabla^2 V = \frac{d^2V}{dz^2} = 0 \longrightarrow V = Az + B$$

When z=0, V = 0 $\longrightarrow$ B=0

When z=d, V = $V_o$ $\longrightarrow$ $V_o$=Ad or A = $V_o$/d

Hence,

$$V = \frac{V_o z}{d}$$

$$\boldsymbol{E} = -\nabla V = -\frac{dV}{dz}\boldsymbol{a}_z = -\frac{V_o}{d}\boldsymbol{a}_z$$

$$\boldsymbol{D} = \varepsilon\boldsymbol{E} = -\varepsilon_o\varepsilon_r\frac{V_o}{d}\boldsymbol{a}_z$$

Since $V_o = 50$ V and d = 2mm,

<u>V = 25z kV,</u> <u>**E** = - 25$\mathbf{a}_z$ kV/m</u>

$$\boldsymbol{D} = -\frac{10^{-9}}{36\pi}(1.5)25\times10^3\boldsymbol{a}_z = \underline{\underline{-332\boldsymbol{a}_z\text{ nC/m}^2}}$$

$$\rho_s = D_n = \underline{\underline{\pm 332 \text{ nC/m}^2}}$$

The surface charge density is positive on the plate at z=d and negative on the plate at z=0.

**Prob. 6.15** From Example 6.8, solving $\nabla^2 V = 0$ when $V = V(\rho)$ leads to

$$V = \frac{V_o \ln \rho/a}{\ln b/a} = V_o \frac{\ln(a/\rho)}{\ln(a/b)}$$

$$\mathbf{E} = -\nabla V = -\frac{V_o}{\rho \ln b/a}\mathbf{a}_\rho = \frac{V_o}{\rho \ln a/b}\mathbf{a}_\rho, \qquad \mathbf{D} = \varepsilon \mathbf{E} = -\frac{\varepsilon_o \varepsilon_r V_o}{\rho \ln b/a}\mathbf{a}_\rho$$

$$\rho_s = D_n = \pm \left.\frac{\varepsilon_o \varepsilon_r V_o}{\rho \ln b/a}\right|_{\rho=a,b}$$

In this case, $V_o$=100 V, b=5mm, a=15mm, $\varepsilon_r = 2$. Hence at $\rho = 10$mm,

$$V = \frac{100 \ln(10/15)}{\ln(5/15)} = \underline{\underline{36.91 \text{ V}}}$$

$$\mathbf{E} = \frac{100}{10\text{x}10^{-3} \ln 3}\mathbf{a}_\rho = \underline{\underline{9.102\mathbf{a}_\rho \text{ kV/m}}}$$

$$\mathbf{D} = 9.102\text{x}10^3 \text{x} \frac{10^{-9}}{36\pi} 2\mathbf{a}_\rho = \underline{\underline{161\mathbf{a}_\rho \text{ nC/m}^2}}$$

$$\rho_s(\rho = 5\text{mm}) = \frac{10^{-9}}{36\pi}(2)\frac{10^5}{5\ln 3} = \underline{\underline{322 \text{ nC/m}^2}}$$

$$\rho_s(\rho = 15\text{mm}) = -\frac{10^{-9}}{36\pi}(2)\frac{10^5}{15\ln 3} = \underline{\underline{-107.3 \text{ nC/m}^2}}$$

**Prob. 6.16**

$$\frac{1}{\rho}\frac{d^2V}{d\phi^2} = 0 \quad \longrightarrow \quad \frac{d^2V}{d\phi^2} = 0 \quad \longrightarrow \quad \frac{dV}{d\phi} = A$$

$$V = A\phi + B$$

$$0 = 0 + B \quad \longrightarrow \quad B = 0$$

$$50 = A\pi/2 \quad \longrightarrow \quad A = \frac{100}{\pi}$$

$$\boldsymbol{E} = -\nabla V = -\frac{1}{\rho}\frac{dV}{d\phi}\boldsymbol{a}_\phi = -\frac{A}{\rho}\boldsymbol{a}_\phi = \underline{\underline{-\frac{100}{\pi\rho}\boldsymbol{a}_\phi}}$$

---

---

**6.29** Show that the resistance of the bar of Figure 6.17 between the vertical ends located at $\phi = 0$ and $\phi = \pi/2$ is

$$R = \frac{\pi}{2\sigma t \ln \dfrac{b}{a}}$$

**6.30** Show that the resistance of the sector of a spherical shell of conductivity $\sigma$, with cross section shown in Figure 6.37 (where $0 \leq \phi < 2\pi$), between its base (i.e., from $r = a$ to $r = b$) is

$$R = \frac{1}{2\pi\sigma(1 - \cos\alpha)}\left[\frac{1}{a} - \frac{1}{b}\right]$$

**6.31** A spherical shell has inner and outer radii $a$ and $b$, respectively. Assume that the shell has a uniform conductivity $\sigma$ and that it has copper electrodes plated on the inner and outer surfaces. Show that

$$R = \frac{1}{4\pi\sigma}\left(\frac{1}{a} - \frac{1}{b}\right)$$

**Prob. 6.29** If the centers at $\phi = 0$ and $\phi = \pi/2$ are maintained at a potential difference of $V_o$, from Example 6.3,

$$E_\phi = \frac{2V_o}{\pi\rho}, \qquad J = \sigma E$$

Hence,

$$I = \int J \bullet dS = \frac{2V_o\sigma}{\pi}\int_{\rho=a}^{b}\int_{z=0}^{t}\frac{1}{\rho}d\rho dz = \frac{2V_o\sigma t}{\pi}\ln(b/a)$$

and

$$R = \frac{V_o}{I} = \frac{\pi}{2\sigma t\ln(b/a)}$$

**Prob. 6.30** If $V(r=a)=0, \quad V(r=b)=V_o$, from Example 6.9,

$$E = \frac{V_o}{r^2(1/a - 1/b)}, \qquad J = \sigma E$$

Hence,

$$I = \int J \bullet dS = \frac{V_o\sigma}{1/a-1/b}\int_{\theta=0}^{\alpha}\int_{\phi=0}^{2\pi}\frac{1}{r^2}r^2\sin\theta d\theta d\phi = \frac{2\pi V_o\sigma}{1/a-1/b}(-\cos\theta)\Big|_0^\alpha$$

$$R = \frac{V_o}{I} = \frac{\frac{1}{a}-\frac{1}{b}}{2\pi\sigma(1-\cos\alpha)}$$

**Prob. 6.31**

This is the same as Problem 6.30 except that $\alpha = \pi$. Hence,

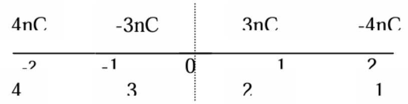
$$R = \frac{1}{2\pi\sigma(1-\cos\pi)}\left(\frac{1}{a}-\frac{1}{b}\right) = \underline{\frac{1}{4\pi\sigma}\left(\frac{1}{a}-\frac{1}{b}\right)}$$

**6.66** Two point charges of 3 nC and −4 nC are placed, respectively, at (0, 0, 1 m) and (0, 0, 2 m) while an infinite conducting plane is at $z = 0$. Determine

(a) The total charge induced on the plane

(b) The magnitude of the force of attraction between the charges and the plane

**Prob. 6.66**




(a) $Q_i$ = -(3nC – 4nC) = $\underline{\underline{\text{1nC}}}$

(b) The force of attraction between the charges and the plates is

$$\mathbf{F} = \mathbf{F}_{13} + \mathbf{F}_{14} + \mathbf{F}_{23} + \mathbf{F}_{24}$$

$$|F| = \frac{10^{-18}}{4\pi \times 10^{-9}/36\pi}\left[\frac{9}{2^2} - \frac{2(12)}{3^2} + \frac{16}{4^2}\right] = \underline{\underline{5.25 \text{ nN}}}$$